

Load Interconnection Request System Impact Study Report

For Queue Project Q19-1343

Toano 345 kV
August 03, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1343. The Interconnection Customer proposes a 394.8 MW large load (data center) facility interconnecting at the Toano 345 kV substation in the Virginia area. The request was received on November 29, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 399 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$28,806,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: Avra Valley / Chuckatuck 138 kV
Contingency: Mira Vista / Yarnen 345 kV
Loading: 121.4% (Rating: 568 MVA)
DFAX: 17.5%
Pre-project loading: 109.2%
2. Facility: Yadkin / No Mans Land 138 kV
Contingency: Mickens / Mira Vista 345 kV
Loading: 117.6% (Rating: 680 MVA)
DFAX: 21.0%
Pre-project loading: 105.4%
3. Facility: Calpine King Energy / Wolcott 138 kV
Contingency: Mira Vista / Yarnen 345 kV
Loading: 116.9% (Rating: 296 MVA)
DFAX: 7.6%
Pre-project loading: 106.7%
4. Facility: Thrasher / Wolcott 138 kV
Contingency: Yarnen / Mira Vista 345 kV
Loading: 114.4% (Rating: 287 MVA)
DFAX: 7.6%
Pre-project loading: 103.9%
5. Facility: Ne-Hobbs / Harbor Beach 138 kV
Contingency: Yarnen / Mira Vista 345 kV Line
Loading: 113.2% (Rating: 411 MVA)
DFAX: 9.6%
Pre-project loading: 103.9%

6. Facility: Harbor Beach / Hickory 138 kV
Contingency: Yarnen / Mira Vista 345 kV line
Loading: 110.9% (Rating: 463 MVA)
DFAX: 9.6%
Pre-project loading: 102.7%
7. Facility: York Canyon Substation / Chico A Substation 138 kV
Contingency: Yrnen / Mira Vista 345 kV
Loading: 106.5% (Rating: 340 MVA)
DFAX: 7.3%
Pre-project loading: 98.0%
8. Facility: Chico A / Northern Neck 138 kV line
Contingency: Yarnen / Mira Vista 345 kV
Loading: 106.4% (Rating: 366 MVA)
DFAX: 7.3%
Pre-project loading: 98.5%
9. Facility: Windsor Dp / Avra Valley 138 kV
Contingency: Mira Vista / Yarnen 345 kV
Loading: 105.2% (Rating: 363 MVA)
DFAX: 9.2%
Pre-project loading: 95.2%
10. Facility: Werner W / Windsor Dp 138 kV
Contingency: Mira Vista / Yarnen 345 kV Line
Loading: 104.3% (Rating: 376 MVA)
DFAX: 9.2%
Pre-project loading: 94.7%
11. Facility: Calpine King Energy 345/138kV Transformer
Contingency: Yarnen / Mira Vista 345 kV
Loading: 103.9% (Rating: 323 MVA)
DFAX: 7.6%
Pre-project loading: 94.5%
12. Facility: Bobville / Avra Valley 138 kV
Contingency: Yarnen / McCartney 345 kV
Loading: 103.5% (Rating: 373 MVA)
DFAX: 9.8%
Pre-project loading: 93.2%
13. Facility: M.L. Hibbard S.E. Station / Travis Ab 138 kV
Contingency: McCartney / Yarnen 345 kV
Loading: 102.6% (Rating: 329 MVA)
DFAX: 6.6%
Pre-project loading: 94.7%
14. Facility: Crittenden / Kraft 138 kV
Contingency: Yarnen / Mira Vista 345 kV Line
Loading: 102.5% (Rating: 600 MVA)
DFAX: 17.5%
Pre-project loading: 91.0%
15. Facility: Kraft / Chuckatuck 138 kV Line
Contingency: Mira Vsta / Yrnen 345 kV line
Loading: 102.0% (Rating: 650 MVA)
DFAX: 17.5%
Pre-project loading: 91.4%
16. Facility: Lightfoot / Toano 138 kV
Contingency: Mira Vista / Yarnen 345 kV

Loading: 100.7% (Rating: 589 MVA)

DFAX: 18.5%

Pre-project loading: 88.3%

17. Facility: Synata Hugoton / Lightfoot 138 kV line

Contingency: Mira Vista / Yarnen 345 kV

Loading: 100.2% (Rating: 631 MVA)

DFAX: 18.5%

Pre-project loading: 88.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Avra Valley / Chuckatuck 138 kV	Rebuild line with higher-capacity conductor	\$4,315,000	18% (\$762,000)
Yadkin / No Mans Land 138 kV	Rebuild line with higher-capacity conductor	\$3,952,000	14% (\$548,000)
Wolcott / Calpine King Energy 138 kV L	Rebuild line with higher-capacity conductor	\$22,827,000	22% (\$4,975,000)
Wolcott / Thrasher 138 kV	Rebuild line with higher-capacity conductor	\$37,474,000	22% (\$8,167,000)
Ne-Hobbs / Harbor Beach 138 kV	Rebuild line with higher-capacity conductor	\$6,827,000	9% (\$601,000)
Hickory / Harbor Beach 138 kV	Rebuild line with higher-capacity conductor	\$7,685,000	9% (\$676,000)
York Canyon Substation / Chico A Substation 138 kV	Reconductor line	\$4,029,000	19% (\$781,000)
Chico A / Northern Neck 138 kV line	Reconductor line	\$804,000	19% (\$156,000)
Windsor Dp / Avra Valley 138 kV	Reconductor line	\$2,802,000	20% (\$551,000)
Werner W / Windsor Dp 138 kV	Reconductor line	\$2,429,000	20% (\$478,000)
Calpine King Energy 345 kV/138 kV Auto	Replace transformer with higher-rated unit	\$30,393,000	22% (\$6,634,000)
Bobville / Avra Valley 138 kV	Reconductor line	\$6,211,000	16% (\$1,014,000)
Travisab / M.L. Hibbard S.E. Sta 138 kV	Reconductor line	\$12,329,000	16% (\$1,984,000)
Crittenden / Kraft 138 kV	Reconductor line	\$2,615,000	18% (\$462,000)
Kraft / Chuckatuck 138 kV Line	Reconductor line	\$3,808,000	18% (\$672,000)
Lightfoot / Toano 138 kV	Reconductor line	\$803,000	14% (\$110,000)
Synata Hugoton / Lightfoot 138 kV line	Reconductor line	\$1,719,000	14% (\$235,000)
	Total allocated to project		\$28,806,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.